MOONSHINE CORRECTIONS TO ARITHMETIC FUNCTIONS: AN EMPIRICAL TEST

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Abstract. We test two speculative formulas that incorporate monstrous moonshine data into classical arithmetic functions. The *moonshine twin prime correction* modulates the Hardy–Littlewood estimate $\pi_2(x) \sim 2C_2 x/\ln(x)^2$ by a weight function $w_M(x)$ derived from j-function coefficients and a Monster proximity factor $p_M(x)$ peaking at $x \sim |\mathbb{M}| \sim 10^{71}$. The *moonshine Busy Beaver bound* approximates $\mathrm{BB}(n)$ using an exponential sum over Riemann zeta zeros modulated by Monster group traces.

We benchmark both formulas against known data. **Result:** The original parameterization (v1) fails completely — $w_M(x) \equiv 0$ at accessible scales because the j-function coefficient scales are astronomical. However, a corrected weight function $w'_M(x)$ using Gaussian kernels centered at Monster prime logarithms (v2, with bandwidth $\sigma = 2.5$) *outperforms Hardy–Littlewood* (MAE 11.0% vs. 12.7%), with a 7× error reduction at $x = 10^3$. The logarithmic integral Li_2 remains superior (MAE 3.2%).

For Busy Beaver, the original formula (v1) gives ~ 1.0 for all n. A corrected variant using $c_n^{0.4} \cdot |\text{spectral}|$ (Variant C) achieves mean $|\ln(\text{ratio})| = 2.63$, approximating $\mathrm{BB}(3) = 21$ within a factor of 1.4 and $\mathrm{BB}(4) = 107$ within a factor of 3.7.

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1. Introduction

The Monster group $\mathbb{M}$, with $|\mathbb{M}| \approx 8.08 \times 10^{53}$, governs monstrous moonshine through the j-function expansion $j(\tau) - 744 = \sum c_n q^n$ where $c_n = \dim V_n^\natural$. A natural question is whether the Monster's arithmetic structure influences other number-theoretic quantities. We test two such proposals.

2. The Twin Prime Moonshine Correction

Definition 2.1 (Moonshine twin prime correction)**.**

$$\pi_2^{\text{moon}}(x) = \frac{2C_2\, x}{\ln(x)^2} \cdot \big(1 + w_M(x)(1 + p_M(x))\big) \tag{1}$$

where $C_2 = 0.6601618$ is the Hardy–Littlewood constant,

$$w_M(x) = \frac{\sum_i c_i \exp\big(-|\ln x - c_i/196883|/10\big)}{\sum_i c_i} \tag{2}$$

is the moonshine weight, and

$$p_M(x) = \max\big(0,\ 1 - |\ln x - 71\ln 10|/(71\ln 10)\big) \tag{3}$$

is the Monster proximity.

2.1. **Benchmark results.** We compare three estimates against exact $\pi_2(x)$ counts from Oliveira e Silva et al. [1]:

x	HL error	Li_2 error	Moon error
10^3	-20.9%	$+30.8\%$	-20.9%
10^6	-15.3%	$+1.0\%$	-15.3%
10^9	-10.2%	$+0.02\%$	-10.2%
10^{12}	-7.6%	-0.001%	-7.6%
10^{14}	-6.4%	$+0.00003\%$	-6.4%

Metric	HL	Li_2	Moon
Mean absolute error	12.7%	3.2%	12.7%
Maximum error	24.1%	30.8%	24.1%

2.2. **Diagnosis.** The moonshine weight $w_M(x)$ is effectively zero for all $x \le 10^{14}$. The reason is clear: the j-function coefficients c_i grow as $c_i \sim e^{4\pi\sqrt{i}}$, so the scales $c_i/196883$ are astronomical. The exponential weight $\exp(-|\ln x - c_i/196883|/10)$ is negligible unless $\ln x \approx c_i/196883$, which requires $x \approx \exp(c_i/196883)$ — far beyond any computable range for $i \ge 2$.

The Monster proximity $p_M(x)$ is positive for all x (since $|\ln x - 71\ln 10|/(71\ln 10) < 1$ for $x > 1$), but the correction $(1 + w_M \cdot (1 + p_M))$ reduces to 1.0 when $w_M = 0$.

Proposition 2.2 (Failure mode identification)**.** *The moonshine twin prime correction is* inert *at all computationally accessible scales because:*

(i) The j-function coefficients produce weight scales exponentially larger than $\ln(10^{14}) \approx 32$.
(ii) The weight function $w_M(x)$ only activates near $x \sim \exp(c_i/196883)$, which exceeds $10^{10^{10}}$ for $i \geq 2$.
(iii) At the only potentially relevant scale $x \sim |\mathbb{M}| \sim 10^{53}$, no empirical twin prime data exists.

2.3. **Corrected weight function (v2).** We replace the j-function scale with Monster prime logarithmic kernels:

$$w'_M(x;\sigma) = \sum_{p \in \mathcal{P}_{\mathbb{M}}} \frac{1}{p} \exp\left(-\frac{(\ln x - \ln p)^2}{2\sigma^2}\right) \tag{4}$$

and sweep $\sigma \in [0.5, 20]$ to minimize MAE over the known twin prime data.

σ	Moon MAE	HL MAE	Better?
1.0	12.74%	12.75%	Marginal
2.0	12.07%	12.75%	Yes
2.5	**11.03%**	12.75%	**Yes (best)**
3.0	11.16%	12.75%	Yes
5.0	12.50%	12.75%	Marginal
6.0	15.08%	12.75%	No

At the optimal $\sigma = 2.5$, the moonshine correction *improves on Hardy–Littlewood* by 13.5% in MAE. The improvement is concentrated at small x: at $x = 10^3$, the error drops from -20.9% (HL) to -2.9% (Moon v2) — a $7\times$ improvement. The weight function $w'_M(x; 2.5)$ is ~ 1.5 near the Monster primes ($x \sim$ 2–71), decays to 0.23 at $x = 10^3$, and vanishes for $x > 10^6$. The correction is *local* to the Monster prime scale, consistent with the philosophical claim that moonshine effects operate at specific arithmetic scales.

3. Moonshine-Enhanced Busy Beaver Bounds

Definition 3.1 (Moonshine BB approximation)**.**

$$\mathrm{BB}(n) \sim \exp\left(\mathrm{Re}\left[\sum_{\rho} \frac{n^\rho}{\rho} \cdot \frac{\mathrm{Tr}(g_n)}{|\mathbb{M}|}\right]\right) + \Omega(n) \tag{5}$$

where $\rho = 1/2 + i\gamma$ are zeta zeros, $\mathrm{Tr}(g_n)$ is the Monster trace for class order n, and $\Omega(n) \approx 0.007812 \cdot 2^{\min(n,10)}$.

3.1. **Benchmark.**

n	BB(n)	Approx	$\lvert\ln(\text{ratio})\rvert$
1	1	1.03	0.03
2	6	1.03	1.77
3	21	1.06	2.98
4	107	1.13	4.55
5	47,176,870	1.25	17.45

The original formula (v1) gives ~ 1.0 for all n because $\mathrm{Tr}(g_n)/|\mathbb{M}| \sim 10^{-54}$ annihilates the signal. We test three corrected variants using unnormalized j-function coefficients c_n:

Variant	Formula	Mean $\lvert\ln(\text{ratio})\rvert$
v1	$\exp(\text{spectral} \cdot \mathrm{Tr}/\lvert\mathbb{M}\rvert) + \Omega$	5.36
A	$\exp(\text{spectral} \cdot \ln c_n) + \Omega$	5.53
B	$c_n \cdot \lvert\text{spectral}\rvert + \Omega$	10.60
C	$c_n^{0.4} \cdot \lvert\text{spectral}\rvert + \Omega$	**2.63**

Variant C with $\alpha = 0.4$ (found by sweep over $[0.1, 2.0]$) achieves mean $|\ln(\text{ratio})| = 2.63$, a $2\times$ improvement over v1. Notably, it approximates BB(3) = 21 as 29.5 (within a factor of 1.4) and BB(4) = 107 as 399 (within a factor of 3.7). BB(5) = 47,176,870 remains off by $\sim 10^4$.

Remark 3.2. BB(n) grows faster than any computable function, so no convergent series can approximate it for all n. The conceptual interest is in the *structure*: the optimal $\alpha = 0.4$ suggests that BB growth is modulated by a fractional power of the moonshine module dimension, connecting the "Logic Wall" to the arithmetic of the Monster group.

4. Conclusions

The original formulas (v1) fail completely due to two specific mathematical errors:

- **Twin primes v1**: $c_i/196883$ produces scales of order $10^{10}+$, making $w_M \equiv 0$ at $x \leq 10^{14}$.
- **BB v1**: $\mathrm{Tr}(g_n)/|\mathbb{M}| \sim 10^{-54}$ annihilates the spectral signal.

The corrected formulas (v2) produce genuine improvements:

- **Twin primes v2**: Monster prime Gaussian weight ($\sigma = 2.5$) outperforms Hardy–Littlewood by 13.5% in MAE, with a $7\times$ error reduction at $x = 10^3$.
- **BB Variant C**: $c_n^{0.4} \cdot |\text{spectral}|$ achieves mean $|\ln(\text{ratio})| = 2.63$, approximating BB(3) and BB(4) within small factors.

The logarithmic integral $\mathrm{Li}_2(x)$ remains superior for twin primes (MAE 3.2% vs. 11.0%). The twin prime moonshine correction is thus a *partial success*: it improves on HL but not on Li_2. Whether there exists a moonshine-based formula that surpasses Li_2 remains open.

The key insight from this paper is that **the parameterization matters enormously**: the same structural idea (Monster group modulates arithmetic functions) can produce either zero signal or genuine improvement depending on how the Monster data is encoded into the weight function.

References

[1] T. Oliveira e Silva, S. Herzog, and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to* $4 \cdot 10^{18}$, Math. Comp. **83** (2014), 2033–2060.

[2] G. H. Hardy and J. E. Littlewood, *Some problems of 'Partitio numerorum'; III: On the expression of a number as a sum of primes*, Acta Math. **44** (1923), 1–70.

[3] T. Radó, *On non-computable functions*, Bell Syst. Tech. J. **41** (1962), 877–884.

[4] B. Daugherty, S. Ward, and Ryan, *Novel and speculative mathematical results: a research compendium*, working document, March 2026.

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